Science practice booklet: Grade 7 expectations

Grade 7 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Science at Grade 7 level, following the Next Generation Science Standards standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Section 1: Living things

Standards practised: MS-LS3-1, MS-LS3-2, MS-LS4-4, MS-LS4-5

A worked example

A group of desert beetles has two colours: pale sand-coloured beetles and dark beetles. Birds find the dark beetles easily against the pale sand and eat more of them. Some beetles in each colour survive and have young. What is likely to happen to the colours of the beetles over many generations, and why?

1. Colour is an inherited trait, so young beetles tend to have the colour of their parents.
2. The pale beetles are hidden better, so more of them survive long enough to reproduce.
3. They pass on the genes for pale colour more often than the dark beetles pass on theirs.
4. Generation after generation, a larger share of the beetles will be pale. This is natural selection.

More and more of the beetles will be pale, because pale beetles survive and reproduce more often and pass on their genes.

Questions

1. What is a gene, and where are genes found?

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2. Which of these is an inherited trait?

   A. Riding a bicycle

   B. Speaking Arabic

   C. The shape of a seed

   D. Knowing the way to school

3. A falcon chick hatches at a falcon centre in the UAE. How many parents gave genes to the chick, and why is the chick not an exact copy of either parent?

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4. What is asexual reproduction? Give one example from plants.

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5. Date palms are either male trees or female trees. Pollen from a male flower must reach a female flower before dates can form. What is this process called, and which type of reproduction is it?

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6. Salem and his sister have the same parents, but they do not look exactly alike. Which statement explains this?

   A. Salem got all of his genes from his father

   B. Salem's genes changed because he is older

   C. They eat different foods at home

   D. Each child gets a different mixture of genes from the same two parents

7. What is a mutation? Can a mutation be harmful, helpful or have no effect?

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8. The Arabian oryx has a pale coat that helps reflect sunlight. Why is the pale coat called an adaptation?

   A. It is an inherited feature that helps the oryx survive in the hot desert

   B. The oryx grew it by standing in the Sun

   C. It changes colour every day to match the sand

   D. Every animal in every habitat has this feature

9. In a group of hares, some have longer legs and can run faster from a fox. What is likely to happen over many generations?

   A. Every hare will grow longer legs by running more

   B. More of the hares in the group will have long legs

   C. Hares with short legs will grow longer legs

   D. Leg length will stay the same because traits are not passed on

10. Fossils show that many living things that lived long ago look different from living things today. What does this suggest about living things over a very long time?

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11. A farmer plants seeds only from the tomato plants that gave the biggest fruit, year after year. What is this called?

    A. Natural selection by predators

    B. A mutation made by the farmer

    C. Selective breeding

    D. Asexual reproduction

12. A long dry spell leaves very little plant food in a desert reserve. Only some of the gazelles in a herd survive. Using the ideas of variation and natural selection, explain why.

Section 2: Earth and space

Standards practised: MS-ESS2-5, MS-ESS2-4, MS-ESS2-6, MS-ESS3-3, MS-ESS3-4, MS-ESS3-5

A worked example

The table gives the average yearly temperature in a city (example data): 1995: 26.8 C, 2005: 27.3 C, 2015: 27.8 C, 2025: 28.4 C. How much did the average temperature rise from 1995 to 2025, and what does this pattern suggest?

1. Subtract the first value from the last: 28.4 - 26.8 = 1.6.
2. Check the pattern: each reading is higher than the one before it.
3. A steady rise over thirty years is a trend, not a single hot year.
4. Scientists look for this kind of long-term trend, across many places, as evidence of a changing climate.

It rose by 1.6 C, a steady upward trend over 30 years.

Questions

1. Which of these is a non-renewable resource?

   A. Sunlight

   B. Wind

   C. Moving water in a river

   D. Natural gas

2. Why is a solar park a good choice for electricity in the UAE?

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3. Burning petrol releases carbon dioxide. Why does this gas matter for the climate?

   A. It cools the air around Earth

   B. It is a greenhouse gas that traps heat in the atmosphere

   C. It blocks all sunlight from reaching Earth

   D. It is the gas that plants give out when they make food

4. Name two pieces of evidence that Earth's climate has been warming in recent decades.

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5. Which statement describes climate rather than weather?

   A. Summers in Abu Dhabi are usually hot and humid

   B. It is cloudy in Sharjah this morning

   C. The temperature in Dubai reached 41 °C today

   D. Fog covered the road at dawn

6. On a hot afternoon at a Dubai beach, Aisha feels a cool wind blowing from the sea to the land. Warm air rises and cooler air sinks. Use this to explain the sea breeze.

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7. After a rain shower, a puddle on a school playground in Ajman disappears on a sunny day. What happened to the water, and where did the energy for this come from?

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8. Which list shows the water cycle in the correct order?

   A. Condensation, evaporation, precipitation

   B. Precipitation, condensation, evaporation

   C. Evaporation, condensation, precipitation

   D. Evaporation, precipitation, condensation

9. Desalination plants give the UAE fresh water, but they use a lot of energy. Give one way to reduce how much water we need to make.

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10. Mangrove trees take in carbon dioxide and protect the coast. Give two reasons why planting more mangroves is a good way to care for the environment.

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11. As more families move into a UAE city, what usually happens to the amount of water, food and energy the city needs? Why?

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12. Which way of getting to work usually puts the least carbon dioxide into the air for each person?

    A. Driving a car alone

    B. Taking a taxi alone

    C. Driving a larger car alone

    D. Riding a full metro train

Section 3: Matter, forces and energy

Standards practised: MS-PS1-1, MS-PS1-2, MS-PS1-5, MS-PS2-1, MS-PS2-2

A worked example

Hamdan seals a fizzy tablet and some water inside a strong bag. The bag and its contents have a mass of 150 g. The tablet fizzes and the bag swells with gas. What is the mass of the sealed bag after the reaction, and why?

1. A chemical reaction rearranges atoms into new substances.
2. No atoms are made or destroyed, and nothing can escape from a sealed bag.
3. So the total mass stays the same: it is still 150 g.
4. The gas is still inside the bag. If the bag were open, the gas would escape and the scale would show less.

150 g, because mass is conserved: the atoms are only rearranged and none left the bag.

Questions

1. What is the difference between an atom and a molecule?

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2. A water molecule is written H2O. How many atoms does one molecule have in total?

   A. 2 atoms

   B. 3 atoms

   C. 4 atoms

   D. 1 atom

3. Mariam mixes baking soda and vinegar in the kitchen and sees bubbles. Bubbles of gas are one sign that a chemical reaction may have taken place. Name one other sign.

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4. A fisherman in Fujairah leaves an iron hook on his boat, and it becomes covered in orange-brown rust. Is rust the same substance as iron? How do you know?

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5. 4 g of hydrogen react completely with 32 g of oxygen to make water. What is the mass of the water made?

   A. 36 g

   B. 28 g

   C. 32 g

   D. 128 g

6. A reaction begins with 8 atoms. How many atoms are there after the reaction? Explain.

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7. Noura rides her bicycle 36 km in 3 hours at a steady speed. What is her speed?

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8. A book rests on a table. Gravity pulls it down, yet it does not move. Why?

   A. Gravity does not act on objects at rest

   B. Friction between the book and the table cancels gravity

   C. The table pushes up with a force equal to gravity, so the forces are balanced

   D. The book is too light for gravity to act on

9. A ball rolls across a classroom floor and gradually stops. Which force makes it stop, and are the forces on it balanced?

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10. A bus brakes suddenly and the passengers lean forward. Which statement explains this?

    A. The brakes push the passengers forward

    B. The bus pulls the passengers forward

    C. Gravity pulls the passengers forward

    D. The passengers' bodies tend to keep moving forward as the bus slows

11. Khalid pushes an empty shopping trolley and then pushes the same trolley full of shopping, using the same force. Which one speeds up more, and why?

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12. A rocket pushes hot gas downwards and the rocket moves upwards. Which law does this show?

    A. Newton's third law

    B. Newton's first law

    C. Newton's second law

    D. The law of conservation of mass

Answer guide for parents

Section 1: Living things

1. A gene is a section of DNA that carries instructions for a trait. Genes are found on chromosomes inside the nucleus of a cell.
2. C. The shape of a seed. The shape of a seed is passed on through genes. The other three are learned.
3. Two parents each gave half of the genes, so the chick has a new mixture of genes and is different from both parents.
4. Reproduction with only one parent, so the offspring have the same genes as the parent. Example: a new date palm that grows from an offshoot of a parent palm.
5. Pollination. It is sexual reproduction, because genes from two parents are combined in the seed.
6. D. Each child gets a different mixture of genes from the same two parents. Each child gets a different mixture of the genes from the two parents, so brothers and sisters show variation.
7. A mutation is a change in a gene. It can be harmful, helpful or have no effect on the living thing, depending on the change and the environment.
8. A. It is an inherited feature that helps the oryx survive in the hot desert. An adaptation is an inherited feature that helps a living thing survive in its environment.
9. B. More of the hares in the group will have long legs. Faster hares survive longer and have more young, so over many generations more hares have long legs.
10. Living things change over very long periods of time, and many kinds that once lived are now extinct.
11. C. Selective breeding. It is selective breeding: people choose which plants pass on their genes, so the fruit is likely to get bigger.
12. Individuals in a herd vary. Some may have traits, such as needing less food or water, that suit the new conditions, so they survive and pass on those genes to their young.

Section 2: Earth and space

1. D. Natural gas. Natural gas takes millions of years to form, so it can run out. Sunlight, wind and moving water are replaced naturally.
2. The UAE has a lot of strong sunshine, sunlight is renewable, and solar panels make electricity without burning fuel.
3. B. It is a greenhouse gas that traps heat in the atmosphere. Carbon dioxide is a greenhouse gas, so adding more of it tends to make Earth warmer.
4. Any two of: rising global average temperatures, melting glaciers and ice sheets, rising sea levels, warmer oceans.
5. A. Summers in Abu Dhabi are usually hot and humid. Climate is the usual pattern of weather over many years. The other statements describe one place at one time.
6. The land heats up faster than the sea, so the air over the land warms and rises. Cooler air from over the sea moves in to replace it, and this moving air is the sea breeze.
7. The water evaporated: it changed into water vapour and rose into the air. The energy came from the Sun.

8. C. Evaporation, condensation, precipitation. Water evaporates, the vapour cools and condenses into clouds, and then falls as precipitation.
9. For example: fix leaks, use less water at home, reuse treated water for irrigation. (Any sensible answer.)
10. They take carbon dioxide from the air, shelter fish and birds, and protect the shore from waves. (Any two.)
11. The need grows, because more people use more resources every day.
12. D. Riding a full metro train. Many people share one train, so less fuel or electricity is used for each person.

Section 3: Matter, forces and energy

1. An atom is a very small particle that makes up matter. A molecule is two or more atoms joined together.
2. B. 3 atoms. There are 2 hydrogen atoms and 1 oxygen atom: 2 + 1 = 3.
3. Any one of: a colour change, a new solid forms, heat or light is given out.
4. No. Rusting is a chemical reaction in which iron reacts with oxygen (and water) to make a new substance with different properties, such as colour and hardness.
5. A. 36 g. 36 g (4 + 32). Mass is conserved in a chemical reaction.
6. 8 atoms. The atoms are rearranged into new substances, but none are made or destroyed.
7. 12 km per hour (36 divided by 3)
8. C. The table pushes up with a force equal to gravity, so the forces are balanced. The upward push of the table equals the downward pull of gravity. The forces are balanced, so there is no change in motion.
9. Friction between the ball and the floor. The forces are unbalanced, so the ball's motion changes.
10. D. The passengers' bodies tend to keep moving forward as the bus slows. Newton's first law: an object keeps moving at the same speed in a straight line unless an unbalanced force acts on it.
11. The empty trolley. For the same force, an object with less mass speeds up more.
12. A. Newton's third law. Newton's third law: when one object pushes on another, the second pushes back on the first with an equal force in the opposite direction.